

On the γ -positivity of Multiset Eulerian polynomials

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Outline of the talk

Let $A_n^{(p)}(t)$ (resp. $B_n^{(p)}(t)$) be the descent polynomial on permutations of the multiset $\{1^p, 2^p, \dots, n^p\}$ (resp. $\{1^p, 2^p, \dots, n^p, n+1\}$).

Outline of the talk

- Multiset Eulerian polynomials
- γ -positivity and bi- γ -positivity
- The bi- γ -positivity of $B_n^{(p)}(t)$
- Weakly increasing trees and an algorithm for trees
- Combinatorial interpretation for the γ -coefficients of $A_n^{(p)}(t)$.

Multiset Eulerian polynomials

Let $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ be a multiset and $p = p_1 + \dots + p_n$.

Definition

A *descent* of $\pi \in \mathfrak{S}_M$ is an index $i \in [p-1]$ such that $\pi_i > \pi_{i+1}$. Denote by $\text{des}(\pi)$ and the number of descents of π , respectively.

$$\text{des}(31\textcolor{red}{7}21556\textcolor{red}{7}24) = \#\{3, 4, 9\} = 3$$

The *multiset Eulerian polynomial* $A_M(t)$ is defined by

$$A_M(t) = \sum_{\pi \in \mathfrak{S}_M} t^{\text{des}(\pi)}.$$

Theorem (MacMahon 1960)

$$\frac{\textcolor{red}{A}_M(t)}{(1-t)^{p+1}} = \sum_{k \geq 0} \binom{p_1 + k}{p_1} \dots \binom{p_n + k}{p_n} t^k.$$

Symmetric, unimodal and γ -positive

A polynomial $h(t) = \sum_{i=0}^n a_i t^i$ is

- **symmetric** if $a_i = a_{n-i}$ for all i
- **unimodal** if for some c

$$a_0 \leq a_1 \leq \cdots \leq a_c \geq \cdots \geq a_{n-1} \geq a_n$$

Example

The Eulerian polynomial $A_n(t)$ is symmetric and unimodal.

Definition

A symmetric polynomial $h(t)$ can be expanded uniquely as

$$h(t) = \sum_{k=0}^{\lfloor n/2 \rfloor} \gamma_k t^k (1+t)^{n-2k}.$$

If $\gamma_k \geq 0$ for all k , then $h(t)$ is said to be **γ -positive**.

Alternatingly increasing and Symmetric decomposition

A polynomial $\sum_{i=0}^n a_i t^i$ is *alternatingly increasing* (or has spiral property) if

$$a_0 \leq a_n \leq a_1 \leq a_{n-1} \leq \cdots \leq a_{\lfloor \frac{n+1}{2} \rfloor}.$$

Alternatingly increasing is a stronger property than unimodality.

Proposition (Brändén)

Let $h(t)$ be a polynomial of degree n . Then $h(t)$ can be written uniquely as $h(t) = a(t) + tb(t)$, where $a(t)$ is symmetric with degree n and $b(t)$ is symmetric with degree less than n . More precisely, we have

$$a(t) = \frac{h(t) - t^{n+1}h(1/t)}{1-t} \quad \text{and} \quad b(t) = \frac{t^n h(1/t) - h(t)}{1-t}.$$



P. Brändén

Symmetric decompositions and real-rootedness,

Int. Math. Res. Not., 10 (2021), 7764–7798.

Example

The symmetric decomposition of $h(t) = 1 + 24t + 120t^2 + 104t^3 + 16t^4$ is

$$h(t) = 1 + 9t + 25t^2 + 9t^3 + t^4 + t(15 + 95t + 95t^2 + 15t^3).$$

Definition

If $a(t)$ and $b(t)$ are both γ -positive, then we say that $h(t)$ is *bi- γ -positive*. Note that if a polynomial is bi- γ -positive, then it is alternatingly increasing.



C.A. Athanasiadis

Gamma-positivity in combinatorics and geometry,
Sém. Lothar. Combin., 77 (2018), Article B77i.

Main result I

Theorem

Fix $p \geq 1$. Let $B_n^{(p)}(t) := A_M(t)$ with $M = \{1^p, 2^p, \dots, n^p, n+1\}$. Then $B_n^{(p)}(t)$ is bi- γ -positive with symmetric decomposition

$$B_n^{(p)}(t) = a_n(t) + tb_n(t),$$

where

$$a_n(t) = \left(1 + (pn - p + 1)t + t(1 - t)\frac{\partial}{\partial t}\right)A_n^{(p)}(t)$$

$$b_n(t) = (p - 1)A_n^{(p)}(t).$$

Consequently, $B_n^{(p)}(t)$ is alternatingly increasing.



S.-M. Ma

Eulerian polynomials for multipermutations and signed multipermutations,

[arXiv:1907.13082](https://arxiv.org/abs/1907.13082).

Computational proof

Using the symmetric decomposition, we have

$$b_n(t) = \frac{t^{p(n-1)+1}B_n^{(p)}(1/t) - B_n^{(p)}(t)}{1-t}$$

and the fact $b_n(t) = (p-1)A_n^{(p)}(t)$ is equivalent to

$$t^{p(n-1)+1}B_n^{(p)}(1/t) - B_n^{(p)}(t) = (1-t)(p-1)A_n^{(p)}(t).$$

Using **MacMahon's** identity in

$$\frac{B_n^{(p)}(t)}{(1-t)^{pn+2}} = \sum_{k \geq 0} \binom{p+k}{p}^n (k+1)t^k$$

which can be rewritten as

$$B_n^{(p)}(t) = \sum_{k \geq 0} t^k \sum_{j=0}^k (-1)^j \binom{pn+2}{j} \binom{p+k-j}{p}^n (k-j+1).$$

Theorem

A function $f : \mathbb{Z} \rightarrow K$ is a polynomial of degree at most k if and only if $\Delta^{k+1}f(n) = 0$.

$$\Delta^k f(n) = (E - 1)^k f(n) = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} f(n+i).$$

Since

$$\binom{p+k}{p}^n (k+1)$$

is a polynomial in k of degree $pn + 1$ and the $(pn + 2)$ -th difference of a polynomial of degree $\leq pn + 1$ vanishes identically, we have

$$\sum_{j=0}^{pn+2} (-1)^j \binom{pn+2}{j} \binom{p+k-j}{p}^n (k-j+1) = 0.$$

Similar manipulation as previous for $(1-t)(p-1)A_n^{(p)}(t)$, we have

$$\begin{aligned} & t^{pn-p+1}B_n^{(p)}(1/t) - B_n^{(p)}(t) \\ &= \sum_{k \geq 0} t^k \sum_{j=0}^{pn+1-k} (-1)^j \binom{pn+2}{j} \binom{pn+1-k-j}{p}^n (1-p) \\ &= (1-t)(p-1)A_n^{(p)}(t). \end{aligned}$$

Which proves $b_n(t) = (p-1)A_n^{(p)}(t)$. It's easy to get $a_n(t)$ by using the **MacMahon's** identity,

$$a_n(t) = \left(1 + (pn - p + 1)t + t(1-t) \frac{\partial}{\partial t} \right) A_n^{(p)}(t).$$

Combinatorial proof

Let $B_{n,k}^{(p)} = [t^k]B_n^{(p)}(t)$ and $A_{n,k}^{(p)} = [t^k]A_n^{(p)}(t)$. Then, $B_{n,k}^{(p)}$ (resp. $A_{n,k}^{(p)}$) counts the number of permutations of $\{1^p, \dots, n^p, n+1\}$ (resp. $\{1^p, \dots, n^p\}$) with k descents. Since every permutation counted by $B_{n,k}^{(p)}$ can be constructed

- either from a permutation counted by $A_{n,k}^{(p)}$ by inserting $n+1$ into one of its k descent positions or the end;
- or from a permutation counted by $A_{n,k-1}^{(p)}$ by inserting $n+1$ into one of its $pn - k$ non-descent positions or the front,

we have

$$B_{n,k}^{(p)} = (k+1)A_{n,k}^{(p)} + (pn+1-k)A_{n,k-1}^{(p)}.$$

The above recurrence relation can be rewritten as

$$B_{n,k}^{(p)} = (k+1)A_{n,k}^{(p)} + (p(n-1)+2-k)A_{n,k-1}^{(p)} + (p-1)A_{n,k-1}^{(p)}.$$

Let f and g be two real-rooted polynomials with $\deg f = \deg g + 1 = n$. If $\{r_i\}$ and $\{s_j\}$ are all zeros of f and g in nonincreasing order, then we say that g *interlaces* f if

$$r_n \leq s_{n-1} \leq \cdots \leq s_2 \leq r_2 \leq s_1 \leq r_1.$$

Theorem (Wang and Yeh 2015)

Suppose that $f(t)$ and $g(t)$ are two real-rooted polynomials with nonnegative coefficients and g interlaces f . Let

$$F(t) = (at + b)f(t) + t(ct + d)g(t).$$

If $ad \geq bc$, then $F(t)$ has also only real zeros.

$B_n^{(p)}(t)$ alternatingly increasing

- Since

$$a_n(t) = \left(1 + (pn - p + 1)t + t(1 - t) \frac{\partial}{\partial t} \right) A_n^{(p)}(t).$$

Note that $h'(t)$ **interlaces** $h(t)$ for any real-rooted polynomial $h(t)$

- By a result of **Simion**, it is clear that $b_n(t) = (p - 1)A_n^{(p)}(t)$ is real-rooted.

Therefore we prove that $B_n^{(p)}(t)$ is alternatingly increasing.



Y. Wang

Polynomials with real zeros and Polya frequency sequences,
J. Combin. Theory Ser. A, **109** (2015) 63-74.



R. Simon

A multi-indexed Sturm sequence of polynomials and unimodality of certain combinatorial sequences,
J. Combin. Theory Ser. A, **36** (1984), 15-22.

Weakly increasing trees on a multiset

Definition (Weakly increasing trees)

Let $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ be a multiset and $p = p_1 + \dots + p_n$. A *weakly increasing tree* on M is a plane tree such that

- 1 it contains $p + 1$ nodes that are labeled by elements from the multiset $M \cup \{0\}$,
- 2 the root is labeled 0 and each node receives a label weakly greater than its parent,
- 3 the labels of the children of each node is weakly increasing from left to right.

Denote by $\mathcal{T}_M$ the set of weakly increasing trees on M .

Weakly increasing trees on $\{1, 2, \dots, n\}$ are *increasing trees*, while Weakly increasing trees on $\{1^n\}$ are *plane trees*.

Weakly increasing trees on a multiset

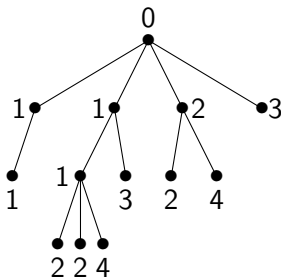


Figure: A weakly increasing tree on $M = \{1^4, 2^4, 3^2, 4^2\}$.

M-polynomial's γ -positivity

A leaf or an internal node (other than the root) of a weakly increasing tree is *old* if it is the leftmost child of its parent; otherwise, it is *young*.

Theorem (Lin–Ma–Ma–Zhou 2021)

For a fixed multiset $M = \{1^{p_1}, 2^{p_2}, \dots, n^{p_n}\}$ with $p = p_1 + \dots + p_n$, the M-Eulerian–Narayana polynomial has γ -positivity expansion

$$\sum_{T \in \mathcal{T}_M} t^{\text{int}(T)-1} = \sum_{i=0}^{\lfloor \frac{p-1}{2} \rfloor} \gamma_{M,i} t^i (1+t)^{p-1-2i},$$

where $\gamma_{M,i}$ enumerates weakly increasing trees on M with $i+1$ leaves and without young leaves.



Z. Lin

Weakly increasing trees on a multiset,

Adv. in Appl. Math., **129** (2021), 102206

Bijection multipermutations and multiset trees

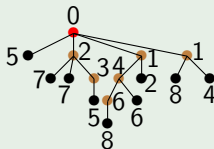
A weakly increasing tree is called a *multiset tree* if no child has the same label as its parent. Let $\mathcal{P}_M$ be the set of all multiset trees in $\mathcal{T}_M$.

Theorem (Lin–Ma–Ma–Zhou 2021)

There exists a natural bijection $\Psi : \mathcal{P}_M \rightarrow \mathfrak{S}_M$ such that $\text{int}(T) = \text{asc}(\Psi(T)) + 1$ for each $T \in \mathcal{P}_M$.

Example

If $\pi = 527735146862184$



when $p_1 = p_2 = \cdots = p_n = 2$, Lin et al. give the combinatorial interpretation for the γ -coefficients of $A_n^{(2)}(t)$.

Theorem

Fix an integer $p \geq 1$. For any $n \geq 1$,

$$A_n^{(p)}(t) = \sum_{i=0}^{\lfloor \frac{p(n-1)}{2} \rfloor} \gamma_{n,i}^{(p)} t^i (1+t)^{p(n-1)-2i},$$

where $\gamma_{n,i}^{(p)}$ is the number of weakly increasing trees on $\{1^p, 2^p, \dots, (n-1)^p, n\}$ with $i+1$ leaves and without young leaves.

Equidistribution

Let $\mathcal{P}_n^{(p)}$ be the set of multiset trees on $\{1^p, 2^p, \dots, n^p\}$.

$$A_n^{(p)}(t) = \sum_{T \in \mathcal{P}_n^{(p)}} t^{\text{int}(T)-1}.$$

Let $\mathcal{T}_n^{(p)}$ be the set of weakly increasing trees on $\{1^p, 2^p, \dots, (n-1)^p, n\}$. we have the following equidistribution

Theorem (Equidistribution)

For integers $p, n \geq 1$,

$$\sum_{T \in \mathcal{P}_n^{(p)}} t^{\text{int}(T)} = \sum_{T \in \mathcal{T}_n^{(p)}} t^{\text{int}(T)}.$$

Degree types of trees

For a tree $T \in \mathcal{T}_M$ with $M = \{1^{p_1}, \dots, n^{p_n}\}$ and a letter $k \in [n]$, we say that k has degree type $\mathbf{t}_k = (t_1, t_2, \dots, t_{p_k})$ if there are t_i ($1 \leq i \leq p_k$) nodes that have exactly i children labeled by k in T .

- a multiset tree $T \in \mathcal{P}_n^{(p)}$ has (letter-degree) type $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ if the letter $i+1$ has degree type $\mathbf{t}_i$ in T for $1 \leq i \leq n-1$.
- a weakly increasing tree $T \in \mathcal{T}_n^{(p)}$, we say T has (letter-degree) type $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ if the letter i has degree type $\mathbf{t}_i$ in T for $1 \leq i \leq n-1$.

Example and refined theorem

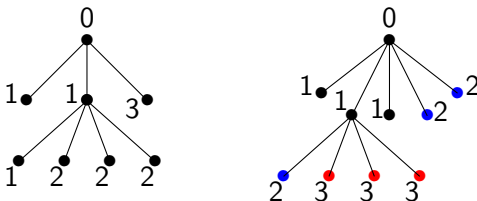


Figure: (Left) A weakly increasing tree in $\mathcal{T}_3^{(3)}$ and (Right) a multiset tree in $\mathcal{P}_3^{(3)}$ with 2 internal nodes and type $((1, 1, 0), (0, 0, 1))$.

Theorem (Refined Theorem by degree types)

For any integer k and type $\mathbf{t}$, there are equal number of multiset trees in $\mathcal{P}_n^{(p)}$ and weakly increasing trees in $\mathcal{T}_n^{(p)}$ with k internal nodes and type $\mathbf{t}$.

Multiset trees recurrence relation for $G_{\mathbf{t},k}$

Denote by $\mathfrak{G}_{\mathbf{t},k}$ (resp. $\mathcal{H}_{\mathbf{t},k}$) the set of trees in $\mathcal{P}_n^{(p)}$ (resp. $\mathcal{T}_n^{(p)}$) with k internal nodes and type $\mathbf{t}$. Let $G_{\mathbf{t},k} = |\mathfrak{G}_{\mathbf{t},k}|$ and $H_{\mathbf{t},k} = |\mathcal{H}_{\mathbf{t},k}|$, a rooted tree of height one will be called a *small tree*.

Lemma

Let $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ where $\mathbf{t}_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1}, \mathbf{t}_n)$ where $\mathbf{t}_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + pn_p = p$. Then

$$G_{\mathbf{t}',k} = \sum_{l \leq k} G_{\mathbf{t},l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \binom{m}{n_1, n_2, \dots, n_p}.$$

Since there are $\binom{pn+1-l}{k-l}$ ways to choose $k-l$ leaves, $\binom{l}{m-k+l}$ to choose $m-k+l$ internal nodes and $\binom{m}{n_1, n_2, \dots, n_p}$ ways to attach these m small trees.

Chen's general bijective algorithm for trees

His bijection between forests of k [small trees](#) on $[n + k - 1]$ and [Schröder trees](#) on $[n]$ with k blocks can be described in the following steps. Given such a forest F of k small trees:

- 1 Mark all nodes $n + 1, n + 2, \dots, n + k - 1$ in F by the symbol $*$.
- 2 Find the tree T in F with no marked node and has the smallest root. Let i be the root of T .
- 3 Find the tree T^* in F that contains the smallest marked node. Let j^* be this marked node.
- 4 There are two ways to merge T and T^* called *horizontal merge* and *vertical merge*.
- 5 Repeat the above procedure until F becomes a Schröder trees.



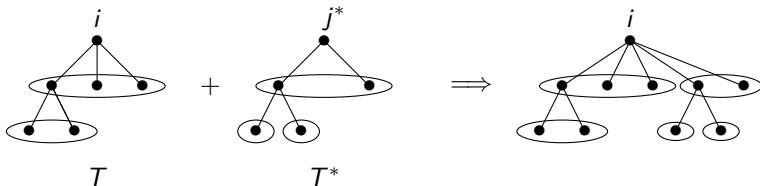
W.Y.C. Chen

A general bijective algorithm for trees,

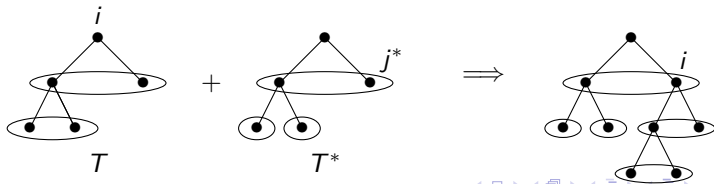
Proc. Natl. Acad. Sci. USA, **87** (1990), 9635–9639.

Horizontal merge and Vertical merge

- i) **Horizontal merge:** If j^* is the root of T^* , then merge T and T^* by identifying i and j^* , keep i as the new node, and put the subtree T^* at the right-hand side of T .



- ii) **Vertical merge:** If j^* is the leaf of T^* , then merge T and T^* by identifying i and j^* and keep i as the new node.



Example

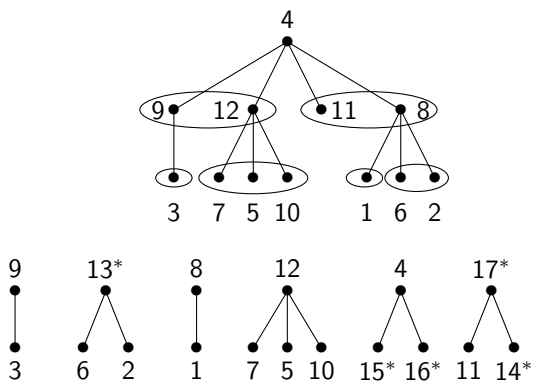


Figure: A Schröder tree on $\{1, 2, \dots, 12\}$ and its small tree decomposition

Counting plane trees with given degree type

Lemma

The number of proper plane trees on $n + i + 1$ nodes with root degree i and degree type $(n_1, n_2, \dots, n_{i-1}, n_i + 1, n_{i+1}, \dots)$ is

$$\frac{i(m-1)!}{n_1!n_2!\cdots} \binom{n}{m-i}$$

whenever $n = \sum_{i \geq 1} in_i$ and $m = \sum_{i \geq 1} n_i$.

Under Chen's bijection, labeled proper plane trees with root degree i and degree type $(n_1, n_2, \dots, n_{i-1}, n_i + 1, n_{i+1}, \dots)$ are in one-to-one correspondence with forests of $m + 1$ small trees such that

- all roots of small trees are unmarked,
- the small tree with greatest marked leaf must have size i and all the other leaves are marked,
- there are exactly $n_i + 1$ small trees of size i and n_j small trees of size j for $j \neq i$.

Weakly increasing trees recurrence relation for $H_{t,k}$

Denote by $\mathcal{A}_{t,k}$ the set of weakly increasing trees with k internal nodes and type $\mathbf{t}$. Let $A_{t,k}$ be the cardinality of $\mathcal{A}_{t,k}$. We have the following recurrence relation for $A_{t,k}$.

Lemma

Let $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ where $\mathbf{t}_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1}, \mathbf{t}_n)$ where $\mathbf{t}_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + pn_p = p$. Then

$$A_{\mathbf{t}',k} = \sum_{l \leq k} A_{\mathbf{t},l} \binom{m}{n_1, n_2, \dots, n_p} \binom{pn+1-l}{k-l} \binom{l}{l+m-k} \left(1 - \frac{p(k-l)}{m(pn+1-l)}\right).$$

The pair decomposition

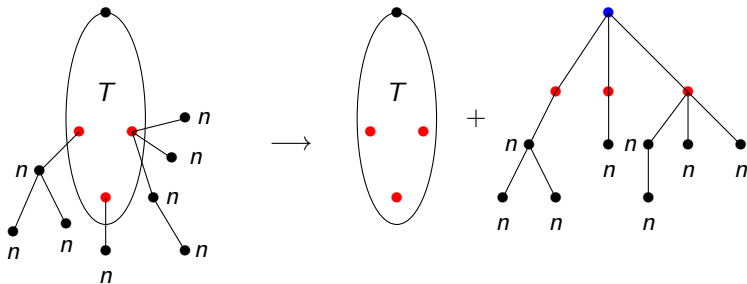


Figure: The pair decomposition of weakly increasing trees.

$$A_{t',k} = \sum_{l \leq k} A_{t,l} \sum_{i=1}^m \frac{i(m-1)!}{n_1! n_2! \cdots n_p!} \binom{p}{m-i} \binom{l}{l+m-k} \binom{(n-1)p+1-l}{k+i-l-m}.$$

Weakly increasing trees recurrence relation for $H_{t,k}$

Lemma

Let $\mathbf{t} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1})$ where $\mathbf{t}_i \in \mathbb{N}^p$. Let $\mathbf{t}' = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_{n-1}, \mathbf{t}_n)$ where $\mathbf{t}_n = (n_1, n_2, \dots, n_p) \in \mathbb{N}^p$ satisfying $n_1 + n_2 + \dots + n_p = m$ and $n_1 + 2n_2 + pn_p = p$. Then

$$H_{\mathbf{t}',k} = \sum_{l \leq k} H_{\mathbf{t},l} \binom{pn+1-l}{k-l} \binom{l}{m-k+l} \binom{m}{n_1, n_2, \dots, n_p}.$$

Since every weakly increasing tree on $\{1^p, 2^p, \dots, n^p, n+1\}$ can be constructed from a unique weakly increasing tree on $\{1^p, 2^p, \dots, n^p\}$ by attaching the node $n+1$, we have

$$H_{\mathbf{t}',k} = kA_{\mathbf{t}',k} + (pn+2-k)A_{\mathbf{t}',k-1}.$$

Similarly, we also have

$$H_{\mathbf{t},l} = lA_{\mathbf{t},l} + (p(n-1)+2-l)A_{\mathbf{t},l-1}.$$

Recurrence for the M -Eulerian–Narayana numbers

For a multiset M , define the M -Eulerian–Narayana number $A_{M,k}$ as the number of weakly increasing trees in $\mathcal{T}_M$ with k internal nodes.

Theorem

Let $M = \{1^{p_1}, 2^{p_2}, \dots, (n-1)^{p_{n-1}}\}$ and $M' = M \cup \{n^{p_n}\}$. Let $p = p_1 + p_2 + \dots + p_{n-1}$ and $p' = p + p_n$. Then

$$A_{M',k} = \sum_{l \leq k} A_{M,l} \binom{p_n + l - 1}{k - 1} \binom{p' + 1 - l}{k - l} \left(1 - \frac{(p_n + l)(k - l)}{k(p' + 1 - l)}\right)$$

with initial condition $A_{\emptyset,l} = \delta_{l,0}$.

- γ -positive has proved,
- **real-rooted** for any M ?
- **two term recurrence relation** for $A_{M,k}$?

Two decompositions of plane trees

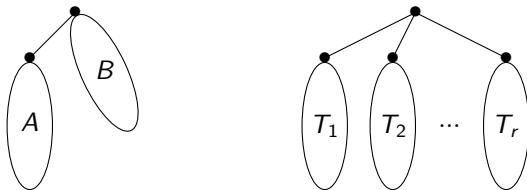


Figure: Two decompositions of plane trees.

Define the generating function

$$N = N(t, x) := \sum_{n \geq 0} x^{n+1} \sum_{T \in \mathcal{T}_n} t^{\text{int}(T)} = x + tx^2 + (t+t^2)x^3 + (t+3t^2+t^3)x^4 + \dots,$$

where $\mathcal{T}_n$ is the set of plane trees with n edges. By the first decomposition of plane trees we have

$$N = x + xtN + N(N - x).$$

Lagrange inversion formula

The compositional inverse of N is

$$N^{\langle -1 \rangle} = \frac{x - x^2}{1 + tx - x}.$$

Let $\mathcal{T}_{n,r}$ denote the set of plane trees with n edges and root degree r .

$$T_{n,r}(t) = \sum_{T \in \mathcal{T}_{n,r}} t^{\text{int}(T)-1}.$$

By the second decomposition of plane trees with root degree r and the *Lagrange inversion formula*, we have (for $r \geq 1$)

$$\begin{aligned} T_{n,r}(t) &= [x^n] N^r \\ &= \frac{r}{n} [x^{n-r}] \left(\frac{1 + x(t-1)}{1-x} \right)^n \\ &= \frac{r}{n} \sum_k \binom{n+k-1}{k} \binom{n}{n-r-k} (t-1)^{n-r-k}. \end{aligned}$$

Triple decomposition of weakly increasing trees

Every weakly increasing tree $T' \in \mathcal{T}_{M'}$ can be decomposed uniquely into a triple (T, P, h)

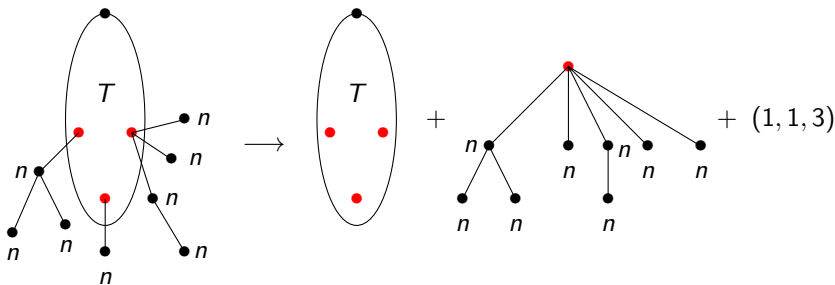


Figure: The triple decomposition of weakly increasing trees.

$$\text{int}(T') = \text{int}(T) + (\text{int}(P) - 1) + \# \text{marked leaves of } T.$$

Sketch of the proof

If P has r branches at the root and T has m marked nodes. For any weakly increasing tree $T \in \mathcal{T}_M$ with l internal nodes, we can

- choose i marked nodes from these l internal nodes of T in $\binom{l}{i}$ ways and choose $m - i$ marked nodes from the remaining $p + 1 - l$ leaves of T in $\binom{p+1-l}{m-i}$ ways;
- from a fixed plane tree $P \in \mathcal{T}_{p_n, r}$, there are $\binom{r-1}{m-1}$ ways to partition the r branches of P into m ordered parts (i.e., to choose a **m -composition** of r in $\binom{r-1}{m-1}$ ways) and then attach these parts to the chosen marked nodes.

Thus, we write

$$t^l \rightsquigarrow \sum_{m, i, r} \binom{l}{i} \binom{p+1-l}{m-i} \binom{r-1}{m-1} T_{p_n, r}(t) t^{l+m-i}$$

this summation can be simplified to

$$\sum_i t^{l+i} \binom{p_n + l - 1}{l + i - 1} \binom{p' + 1 - l}{i} \left(1 - \frac{i(p_n + l)}{(l + i)(p' + 1 - l)}\right)$$

Open problem and Conjecture

- Direct combinatorial proof?
- $|\mathcal{T}_M| = |\mathcal{T}_{M'}|$ with $M = \{1^{2n+2}, 2^n\}$ and $M' = \{1^{2n+1}, 2^{n+1}\}$.

$$|\mathcal{T}_{\{1^x, 2^y\}}| = \frac{1}{1+x+y} \binom{2x}{x} \binom{x+2y}{y}.$$

The solutions for the **Diophantine equation** $f(x, y) = f(x', y')$, where $x, y, x', y' \in \mathbb{P}$, are $(x, y) = (2n+1, n+1)$ and $(x', y') = (2n+2, n)$ for all $n \in \mathbb{P}$. maple verified $\leq 500!$

- Characterize all multisets M and M' for which $|\mathcal{T}_M| = |\mathcal{T}_{M'}|$.
Instance: $M = \{1^2, 2^3\}$ and $M' = \{1^1, 2^1, 3^3\}$

Thanks for you listening !